

SIGNALS AND SYSTEMS TEST 2

Number of Questions: 35

Time: 60 min.

Directions for questions 1 to 35: Select the correct alternative from the given choices.

1. The final value of $X(t)$ is. If $X(s) = \frac{2}{s^2 + 3s + 3}$ is _____.

- (A) 2 (B) zero
(C) 1 (D) ∞

2. If the system is causal and stable, the system poles must lie

- (A) on the $j\omega$ axis
(B) on the left half of s - plane, with negative real part
(C) on the right half of s - plane, with positive real part
(D) both (a) & (b)

3. If the step response of a causal, LTI system is $s(t)$. Then the output of the system is, if the input is step function

- (A) $\int_0^t \frac{ds(\tau)}{d\tau} u(t+\tau) d\tau$
(B) $\int_0^t \frac{ds(\tau)}{d\tau} u(t+\tau) d\tau + u(0)s(t)$
(C) $s(t)$
(D) (a) & (c)

4. The signal $x(t) = t u(t)$ is

- (A) power signal
(B) energy signal
(C) neither energy nor power signal
(D) None of these

5. Invertible systems are those systems where

- (A) input signal can be uniquely determined by observing output signal.
(B) system output is always constant
(C) output can be uniquely obtained from the knowledge of input
(D) A & C both

6. Z and laplace transforms are related by

- (A) $s = \ln z$ (B) $s = \frac{\ln z}{T}$
(C) $s = z$ (D) $s = T \ln z$

7. If $X(\omega)$ is fourier transform of a real signal $x(t)$, then

- (A) $\phi(-\omega) = -\phi(\omega)$ (B) $|X(-\omega)| = |X(\omega)|$
(C) $\phi(-\omega) = \phi(\omega)$ (D) both (a) & (b)

8. The inverse fourier transform of

$$X(\omega) = \frac{1}{2 - \omega^2 + j3\omega}$$

- (A) $[e^{-t} - e^{2t}]u(t)$ (B) $[e^t - e^{2t}]u(t)$
(C) $[e^{-t} - e^{-2t}]u(t)$ (D) None of these

9. If $x[n]$ is real and odd then its discrete Fourier series coefficient C_k will be

- (A) Real (B) even
(C) imaginary (D) B & C

10. If the continuous time signal $X_a(t) = \sin(2500 \pi t)$ is sampled at sampling frequency 10Hz, then the discrete time sequence $x[n]$ is

- (A) $\sin(500 \pi n)$ (B) $\cos(5000 \pi n)$
(C) $\sin(250 \pi n)$ (D) $\cos(250 \pi n)$

11. Nyquist rate for the signal $x(t) = \frac{\sin 160\pi t}{\pi t}$ is _____.

- (A) 160 Hz (B) 320 Hz
(C) 640 Hz (D) None of these

12. If the lower limit of ROC is less than upper limit of

ROC for $X(z) = \sum_{n=-\infty}^{+\infty} x[n]z^{-n}$, then series

- (A) Diverges (B) converges
(C) can't say (D) zero

13. Fundamental period of the signal $x(t) = \cos(2t) + \sin(2\sqrt{2}t)$ is

- (A) π (B) $\frac{\pi}{2}$
(C) Not periodic (D) $\sqrt{2}\pi$

14. For a finite sequence $x[n] = [3, 7, -1, 0, 3, 2]$

The z - transform & ROC is

- (A) $3z^{-2} + 7z^{-1} - 1 + 3z^2 + 2z^3, 0 < |z| < \infty$
(B) $3z^2 + 7z - 1 + 3z^{-2} + 2z^{-3}, 0 < |z| < \infty$
(C) $3z^2 + 7z - 1 + 3z^{-2} + 2z^{-3}, |z| < \infty$
(D) $3z^{-2} + 7z^{-1} - 1 + 3z^2 + 2z^3, |z| > 0$

15. If $x(z)$ has poles at $z = \frac{-1}{3}$ and $z = 1$. Also given that $x[1]$

$$= 1 \text{ and } x[-1] = \frac{1}{2} \text{ and ROC includes the point } z = \frac{2}{3},$$

then the time signal $x[n]$ is _____

- (A) $u[n] + \frac{1}{2} \left(\frac{-1}{3} \right)^{n+1} u[-n-1]$
(B) $u[n] + \frac{1}{6} \left(\frac{-1}{3} \right)^n u[-n-1]$
(C) $u[n] - \frac{1}{2} \left(\frac{-1}{3} \right)^{n+1} u[-n-1]$
(D) None of these

16. If the impulse response of the system is given by $h(t) = e^{-3|t|}$. Then the system is _____

- (A) causal only
(B) Not causal but not “BIBO stable”
(C) causal but unstable
(D) None of these

17. An LTI system is described by

$$H(s) = \frac{1}{s^2 + 3s + 1}.$$

The system response $y(t)$ is, if input $x(t) = 2 \cos(3t + 25^\circ)$

- (A) $0.17 \cos(3t + 73.4^\circ)$ (B) $2 \cos(3t + 48.4^\circ)$
(C) $0.17 \cos(3t + 25^\circ)$ (D) None of these

18. Wrong relationship is

- (A) $h[n] \times u[n] = \sum_{k=-\infty}^{+\infty} h(k)u(n-k)$
(B) $h[n] * x[n] = \sum_{k=-\infty}^{+\infty} h(k)x(n-k)$
(C) $\{x[n] * h_1[n] * h_2[n] = x[n] * \{h_1[n] * h_2[n]\}$
(D) $\sum_{k=-\infty}^{\infty} h(k)x(n-k) = \sum_{k=-\infty}^{\infty} x(k)h(n-k)$

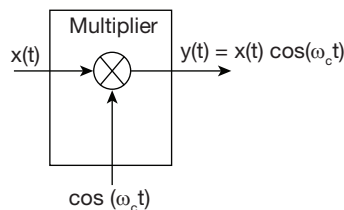
19. The impulse response of a discrete time LTI system is given by

$$h[n] = \left(\frac{1}{3}\right)^n u[n]$$

If $y[n]$ is the output of the system with input $x[n] = \delta[n] + 3\delta[n-4]$ then $y[2]$ and $y[5]$ are

- (A) $\frac{244}{243}$ & $\frac{1}{9}$ (B) $\frac{1}{3}$ & zero
(C) $\frac{244}{9}$ & $\frac{244}{243}$ (D) $\frac{1}{9}$ & $\frac{244}{243}$

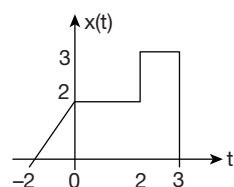
20. Consider the system in figure



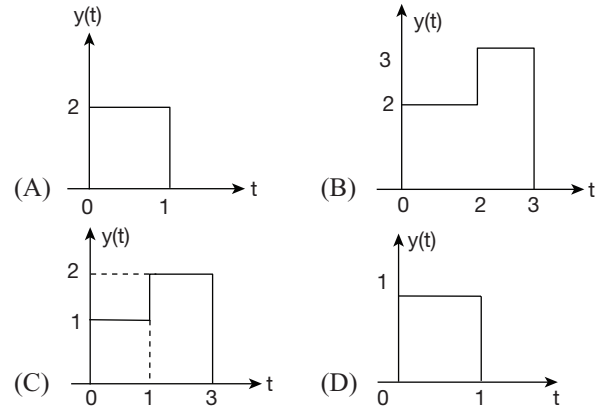
The system is

- (A) memoryless & causal (B) LTI
(C) Stable (D) A & C

21. A continuous time signal $x(t)$ is shown in figure



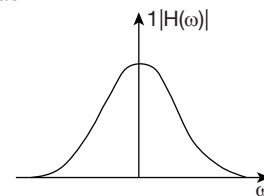
Sketch of the signal $y(t) = x(t)[u(t) - u(t-1)]$ is



22. The equivalent bandwidth of a filter with frequency response $H(\omega)$ is defined by Weq

$$= \frac{1}{|H(\omega)|_{\max}^2} \int_0^{\infty} |H(\omega)|^2 d\omega$$

If the filter is low pass RC filter and magnitude spectrum is given as



Then the equivalent bandwidth Weq is

- (A) $\frac{\pi}{2RC}$ (B) $\frac{\pi}{RC}$
(C) $\frac{\pi}{2(RC)^3}$ (D) None of these

23. In a definition it is given that the bandwidth for a signal $x(t)$ is the 80 % energy containment bandwidth W_{85} , defined by

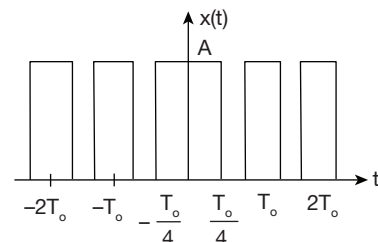
$$\frac{1}{2\pi} \int_{-W_{80}}^{+W_{80}} |x(\omega)|^2 d\omega = \frac{1}{\pi} \int_0^{W_{80}} |x(\omega)|^2 d\omega = 0.8E_x$$

Where E_x is the normalized energy content of the signal $x(t) = e^{-2at}u(t)$, $a > 0$

The W_{80} of the signal $x(t)$ is

- (A) 3.077a (B) 6.15a
(C) 2.33a (D) None of these

24. The Fourier series coefficient C_k of the periodic square wave $x(t)$ is (for odd k value $= 2m+1$)



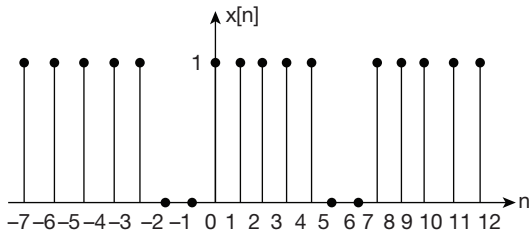
- (A) $(-1)^m \frac{A}{k\pi}$ (B) $\frac{A}{K\pi}$
 (C) $(-1)^m \frac{A}{(2m+1)\pi}$ (D) None of these

25. The discrete fourier series coefficient, C_K for the given signal $x[n] = \cos \frac{\pi}{4} n$ is

- (A) $C_1 = \frac{1}{2}, C_{-1} = -\frac{1}{2}$ & all other $C_K = 0$
 (B) $C_1 = \frac{1}{2}, C_7 = \frac{1}{2}$ & all other $C_K = 0$
 (C) $C_1 = \frac{1}{2}, C_8 = \frac{1}{2}$ & all other $C_K \neq 0$
 (D) None of these

26. If a discrete time LTI system with impulse response

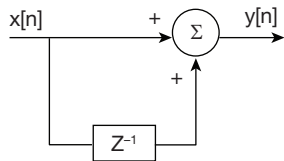
$$h[n] = \frac{\sin\left(\frac{\pi}{8}n\right)}{\pi n}$$



is given then what is output $y[n]$ if the input $x[n]$ is a periodic sequence with fundamental period $N_o = 7$ as shown in figure

- (A) 1 (B) $\frac{3}{5}$
 (C) $\frac{5}{7}$ (D) $\frac{2}{7}$

27. A discrete time LTI System is given in figure.

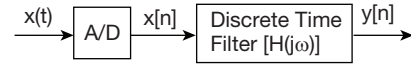


The frequency response $H(\Omega)$ of the system is ____.

- (A) $2e^{-\frac{j\Omega}{2}} \cos\left(\frac{\Omega}{2}\right)$ (B) $e^{-\frac{j\Omega}{2}} \sin\left(\frac{\Omega}{2}\right)$
 (C) $1 + e^{j\Omega}$ (D) Both A & C

28. An A/D converter is cascaded with the discrete time filter having frequency response

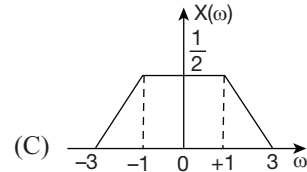
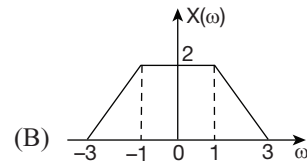
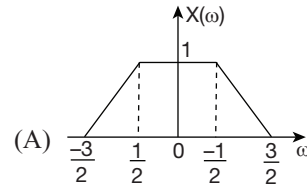
$$H(j\omega) = \frac{1 - e^{-j\omega}}{1 - \frac{1}{2}e^{-j\omega}}$$



If the input $x(t) = \sin(125\pi t)$ and sampling frequency is 500 Hz. Then output $y[n]$ of discrete time filter is

- (A) $\sin\left(\frac{n\pi}{4} - 102.34^\circ\right)$ (B) $\sin\left(\frac{n\pi}{4} - 40^\circ\right)$
 (C) $\cos\left(\frac{n\pi}{4} - 77.64^\circ\right)$ (D) $\cos\left(\frac{n\pi}{4} - 40^\circ\right)$

29. Fourier transform of the signal $x(t) = \frac{\sin 2t \cdot \sin t}{4\pi^2 t^2}$ is



(D) None of these

30. A discrete time signal is given as $x[n] = u[n] - u[n-5]$. The even part of the signal is

- (A) $\frac{u[n] + u[-n-1] - u[n-5] - u[-n+5]}{2}$
 (B) $\frac{u[n] + u[-n] - u[n-5] - u[-n-5]}{2}$
 (C) $\frac{u[n+4] - u[n-5] + \delta[n]}{2}$
 (D) $\frac{1 + \delta[n] - u[n-5] + u[n+4]}{2}$

31. The final value of the signal $x[n]$ is; whose z -transform is given as $X(z) = \frac{z^{-1}(1-z^{-4})}{(1-z^{-1})^2(1+z^{-1})}$

- (A) zero (B) 1
 (C) 2 (D) infinite

Common data for Questions 32 and 33:

A real and odd periodic signal $x(t)$ with period 2 has Fourier series coefficients $X[k] = 0$ for $|k| > 1$ and $\frac{1}{2} \int_0^2 |x(t)|^2 dt = 1$

32. Then the Fourier series coefficient at $k = 0$ is _____

- (A) zero (B) 1
(C) $\frac{1}{2}$ (D) $\frac{1}{3}$

33. The signal $x(t)$ is

- (A) $-\sqrt{2} \sin(\pi t)$ (B) $+\sqrt{2} \sin(\pi t)$
(C) $-\frac{1}{\sqrt{2}} \cos(\pi t)$ (D) $+\frac{1}{\sqrt{2}} \sin(\pi t)$

Statement for linked Answer Questions 34 and 35:

The system formed by connecting two systems in cascade.

The impulse response of the systems are given by $h_1(t) = e^{-3t} u(t)$ and $h_2(t) = 3e^{-t} u(t)$

34. The impulse response $h(t)$ of the over all system is given by _____

- (A) $[e^{-3t} + 3e^{-t}] u(t)$ (B) $\frac{3}{2}[e^{-3t} - e^{-t}] u(t)$
(C) $\frac{3}{2}[e^{-t} - e^{-3t}] u(t)$ (D) None of these

35. The over all system is

- (A) stable (B) unstable
(C) causal (D) Both (A) & (C)

ANSWER KEYS

1. B	2. B	3. C	4. C	5. D	6. B	7. D	8. C	9. C	10. C
11. A	12. B	13. C	14. B	15. A	16. D	17. A	18. A	19. D	20. D
21. A	22. A	23. B	24. A	25. B	26. C	27. D	28. A	29. C	30. C
31. C	32. A	33. A	34. C	35. D					

HINTS AND EXPLANATIONS

1. As we know that final value theorem is

$$\lim_{t \rightarrow \infty} x(t) = \lim_{s \rightarrow 0} s X(s)$$

$$= \lim_{s \rightarrow 0} s \cdot \frac{2}{s^2 + 3s + 3}$$

$$= \lim_{s \rightarrow 0} \frac{2s}{s^2 + 3s + 3} = 0$$

Choice (B)

2. If the system is causal and stable, both then the poles will lie only on the left half of the s - plane & for stability ROC must contain the $j\omega$ axis that is ROC is of the form $\text{Re}(s) > \sigma_{\max}$

Choice (B)

3. If the step response of a causal, LTI system is $s(t)$, then

$$\text{Impulse response } h(t) = \frac{ds(t)}{dt}$$

$$\text{And input } x(t) = u(t)$$

$$\text{So output response } y(t) = \frac{ds(t)}{dt} * u(t)$$

$$= \int_{-\infty}^t \frac{ds(\tau)}{d\tau} u(t-\tau) d\tau = \int_0^t \frac{ds(\tau)}{d\tau} d\tau = s(t)$$

Choice (C)

$$4. \text{ Energy } = E = \lim_{T \rightarrow \infty} \int_{-\frac{T}{2}}^{+\frac{T}{2}} |x(t)|^2 dt$$

$$= \lim_{T \rightarrow \infty} \int_0^T t^2 dt = \lim_{T \rightarrow \infty} \frac{\left(\frac{T}{2}\right)^3}{3} = \infty$$

$$\text{Power } P = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-\frac{T}{2}}^{+\frac{T}{2}} |x(t)|^2 dt$$

$$= \lim_{T \rightarrow \infty} \frac{1}{T} \frac{\left(\frac{T}{2}\right)^3}{3} = \infty$$

So $x(t)$ is neither energy non power signal

Choice (C)

5. Choice (D)

6. If $x(t)$ is sampled at a rate $\frac{1}{T}$ then $x(t)$ can be represent as

$$x(t) = \sum_{k=-\infty}^{+\infty} x(KT) \delta(t - KT) \quad \text{--- (1)}$$

By taking Laplace Transform of equation" (1)

$$X(s) = \sum_{k=-\infty}^{+\infty} r(KT) e^{-KsT}$$

If $e^{+sT} = Z$ then

$$X(s) = \sum_{k=-\infty}^{+\infty} r(K) z^{-K} = X(z)$$

So $Z = e^{sT}$

$$\ln z = sT$$

$$s = \frac{\ell n z}{T}$$

Choice (B)

7. Since $x(t)$ is real function so

By duality property-

$$\text{If } x(t) \leftrightarrow X(\omega)$$

$$X(t) \leftrightarrow 2\pi x(-\omega)$$

$$\text{and the magnitude of } |X(-\omega)| = |X(\omega)|$$

and phase $\phi(\omega) = \tan^{-1}$ (function of ω)

$$\text{so } \phi(\omega) = -\phi(-\omega)$$

$$\text{or } \phi(-\omega) = -\phi(\omega)$$

Choice (D)

$$8. X(\omega) = \frac{1}{2 - \omega^2 + j3\omega} = \frac{1}{2 + (j\omega)^2 + j3\omega}$$

$$= \frac{1}{(1 + j\omega)(2 + j\omega)} = \frac{1}{(1 + j\omega)} - \frac{1}{(2 + j\omega)}$$

By taking inverse fourier transform of $x(\omega)$

$$x(t) = [e^{-t} - e^{-2t}]u(t) \quad \text{Choice (C)}$$

9. If $x[n]$ is real & even then its discrete Fourier series coefficient is real
 $\Rightarrow$ If $x[n]$ is real and odd then its discrete Fourier series coefficient is imaginary Choice (C)

10. $x(t) = \sin(2500\pi t)$

$$x[n] = x[nT_s] = \sin\left(\frac{2500\pi n}{f_s}\right)$$

Where $f_s = \frac{1}{T_s} = 10\text{Hz}$

$$\text{So } x[n] = \sin\left(\frac{2500\pi n}{10}\right) = \sin(250\pi n)$$

Choice (C)

11. $\omega_m = 160\pi$

So $\omega_s = 2 \times 160\pi = 320\pi = 320\pi$

$f_s = 160\text{Hz}$

Choice (A)

12. ROC is defined as range of values for which Z – transform converges Choice (B)

13. $x(t) = \cos(2t) + \sin(2\sqrt{2}t)$

Here $x_1(t) = \cos 2t$, is periodic with period

$$T_1 = \frac{2\pi}{2} = \pi$$

And $x_2(t) = \sin(2\sqrt{2}t)$ is periodic with period

$$T_2 = \frac{2\pi}{2\sqrt{2}} = \frac{\pi}{\sqrt{2}}$$

Since $\frac{T_1}{T_2} = \frac{\pi}{\pi\sqrt{2}} = \sqrt{2}$ is not a rational number

So $x(t)$ is not a periodic signal

Choice (C)

14. $X[z] = \sum_{n=-\infty}^{+\infty} x[n]z^{-n} = \sum_{n=-2}^{+3} x[n]z^{-n}$

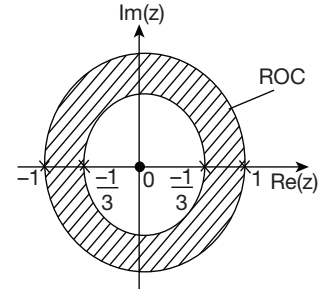
$$= 3z^{+2} + 7z^{+1} - 1 + 3z^{-2} + 2z^{-3}$$

Since both positive & negative powers of z are included so ROC of $X(Z)$ will be $0 < |z| < \infty$

Choice (B)

15. Since ROC includes the point $z = \frac{2}{3}$

So ROC will be $\frac{1}{3} < |z| < 1$



and $X(z)$ has two poles at $z = -\frac{1}{3}$ and $z = 1$

$$\text{so } X(z) = \left[\frac{A}{\left(1 + \frac{1}{3}z^{-1}\right)} + \frac{B}{(1 - z^{-1})} \right]$$

$$\left[\begin{array}{cc} |z| > \frac{1}{3} & |z| < 1 \\ \text{Left sided signal} & \text{Right sided signal} \end{array} \right]$$

$$\text{so } x[n] = -A\left(\frac{-1}{3}\right)^n u[-n-1] + B[1]^n u[n]$$

As given that $x[1] = 1 \Rightarrow 1 = B$

$$\text{and } x[-1] = -A\left(\frac{-1}{3}\right)^{-1} \cdot 1$$

$$\frac{1}{2} = 3A \Rightarrow A = \frac{1}{6}$$

$$\text{and } x[n] = u[n] - \frac{1}{6}\left(\frac{-1}{3}\right)^n u[-n-1]$$

$$= u[n] + \frac{1}{2}\left(\frac{-1}{3}\right)^{n+1} u[-n-1] \quad \text{Choice (A)}$$

16. $h(t) = e^{-3|t|}$

$$h(t) = e^{-3t}u(t) + e^{3t}u(-t)$$

we can see that $e^{3t}u(-t) \neq 0$ for $t < 0$, so given system is not causal.

For checking the system to be “BIBO stable”

We know that $\int_{-\infty}^{+\infty} h(t) dt < \infty$

$$\text{So } \int_{-\infty}^{+\infty} h(t) dt = \int_{-\infty}^0 e^{3t}u(t) dt + \int_0^{+\infty} e^{3t}u(t) dt$$

$$= -\frac{1}{3}[e^{-3t}]_0^{\infty} + \frac{1}{3}[e^{3t}]_{-\infty}^0 = \frac{1}{3}[0 - 1] + \frac{1}{3} = \frac{2}{3} < \infty$$

So the system is “BIBO Stable”

Choice (D)

17. $x(t) = 2 \cos(3t + 25^\circ)$ can be expressed as real part of complex exponential with $S = 3j$

$$x(t) = 2\text{Re}(e^{j(3t+25^\circ)})$$

$$\begin{aligned} \text{So } H(3j) &= \frac{1}{(3j)^2 + (3 \times 3j) + 1} \\ &= \frac{1}{-9 + 9j + 1} = \frac{1}{-8 + 9j} \end{aligned}$$

$$\text{so } = 0.083 e^{j48.4^\circ} = 0.083 e^{j48.4^\circ}$$

$$\begin{aligned} \text{so } Y_{ss}(t) &= 2|H(3j)| \cos(3t + 25^\circ + \angle H(3j)) \\ &= 2 \times 0.083 \cos(3t + 25^\circ + 48.4^\circ) \\ &= 0.17 \cos(3t + 73.4^\circ) \end{aligned}$$

Choice (A)

$$18. h[n] \times u[n] = \sum_{k=-\infty}^{+\infty} h(k)u(n-k) \quad \text{Choice (A)}$$

$$19. y[n] = x[n] * h[n] = (\delta[n] + 3\delta[n-4]) * \left(\frac{1}{3}\right)^n u[n]$$

$$y[n] = \left(\frac{1}{3}\right)^n u[n] + 3\left(\frac{1}{3}\right)^{n-4} u[n-4]$$

$$\text{So } y[2] = \left(\frac{1}{3}\right)^2 u[2] + 3\left(\frac{1}{3}\right)^{-2} u[-2]$$

$$\text{and } y[5] = \left(\frac{1}{3}\right)^5 u[5] + 3\left(\frac{1}{3}\right)^1 u[1] = \left(\frac{1}{3}\right)^5 + 1 = \frac{244}{243}$$

Choice (D)

20. For memory less $y(t) = T\{x(t)\} = x(t)\cos \omega_c t$
Since the value of output $y(t)$ depends on only the present values of the input $x(t)$, The system is memory less

$\Rightarrow$ For causality

Since $y(t)$ does not depend upon future values of the input $x(t)$, so system is causal

$\Rightarrow$ for LTI

$Y_1(t)$ be output produced by the shifted input $x_1(t) = x(t - t_0)$.

Then $y_1(t) = T\{x(t - t_0)\} = x(t - t_0)\cos \omega_c t$

But $y(t - t_0) = x(t - t_0)\cos \omega_c(t - t_0) \neq y_1(t)$

So system is not LTI

$\Rightarrow$ for stability since $\cos \omega_c t \leq 1$

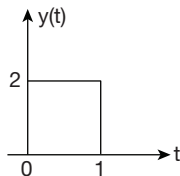
$$|y(t)| = |x(t)\cos \omega_c t| \leq |x(t)|$$

Thus if the input $x(t)$ is bounded then output $y(t)$ is also bounded. So system is BIBO stable

Choice (D)

$$21. \text{ As we know that } u(t) - u(t-1) = \begin{cases} 1 & 0 < t \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

$$\text{So } x(t)[u(t) - u(t-1)] = \begin{cases} 2 & 0 < t \leq 1 \\ 0 & \text{otherwise} \end{cases}$$



Choice (A)

22. for a low pass RC filter

$$H(\omega) = \frac{1}{1 + j\omega RC} = \frac{1}{RC} \frac{1}{1/RC + j\omega}$$

As we can see that maximum value as $|H(\omega)|_{\max} = 1$ at $\omega = 0$

$$\text{So } W_{eq} = \frac{1}{1} \int_0^{\infty} \frac{1}{(RC)^2} \frac{1}{\left(\frac{1}{RC}\right)^2 + (\omega)^2} d\omega$$

$$W_{eq} = \frac{1}{(RC)^2} \int_0^{\infty} \frac{d\omega}{\left(\frac{1}{RC}\right)^2 + \omega^2}$$

$$= \frac{RC}{(RC)^2} \left[\tan^{-1} \omega RC \right]_0^{\infty}$$

$$= \frac{\pi}{2RC}$$

Choice (A)

23. $x(t) = e^{-2at} u(t)$

$$\text{So } X(\omega) = \frac{1}{2a + j\omega}$$

$$\text{Energy } E = \int_{-\infty}^{+\infty} |X(\omega)|^2 d\omega$$

$$\int_0^{\infty} e^{-4at} dt = \frac{1}{4a}$$

Now by the given equation

$$\frac{1}{\pi} \int_0^{W_{80}} |x(\omega)|^2 d\omega = \frac{1}{\pi} \int_0^{W_{80}} \frac{d\omega}{4a^2 + \omega^2}$$

$$\frac{1}{2\pi a} \tan^{-1} \left[\frac{W_{80}}{2a} \right] = 0.8Ex$$

$$\Rightarrow \tan^{-1} \left[\frac{W_{80}}{2a} \right] = \frac{0.8 \times 2\pi a}{4a} = 0.40\pi$$

$$\text{So } W_{80} = 2a \tan(0.4\pi) = 2a \times 3.077 = 6.15a \text{ rad/s}$$

Choice (B)

$$24. x(t) = \sum_{k=-\infty}^{\infty} C_k e^{jk\omega_o t} \quad \omega_o = \frac{2\pi}{T_o}$$

$$C_k = \frac{1}{T_o} \int_{-T_o/2}^{T_o/2} x(t) e^{-jk\omega_o t} dt$$

$$= \frac{A}{-jk\omega_o T_o} \times \left(e^{-jk\omega_o T_o/4} - e^{jk\omega_o T_o/4} \right)$$

$$= \frac{A}{-jk2\pi} \times \left[e^{-jk\pi/2} - e^{jk\pi/2} \right]$$

$$= \frac{A}{k\pi} \sin\left(\frac{k\pi}{2}\right)$$

$$\text{Thus } C_K = 0, \quad k = 2m \neq 0$$

$$C_k = (-1)^m \frac{A}{K\pi} \quad k = 2m + 1$$

$$C_o = \frac{1}{T_o} \int_0^{T_o} x(t) dt = \frac{1}{T_o} \int_0^{T_o/2} A dt = \frac{A}{2}$$

$$\text{Hence } C_o = \frac{A}{2}, C_{2m} = 0, m \neq 0$$

$$C_{2m+1} = (-1)^m \frac{A}{(2m+1)\pi}$$

$$x(t) = \frac{A}{2} + \frac{A}{\pi} \sum_{m=-\infty}^{+\infty} \frac{(-1)^m}{2m+1} e^{j(2m+1)\omega_o} \quad \text{Choice (A)}$$

25. The fundamental period of $x[n]$ is No

$$\text{So, } \Omega_o = \frac{2\pi}{N_o} = \frac{\pi}{4}$$

So $N_o = 8$

By using Euler's formula

$$\cos \frac{\pi}{4} n = \frac{1}{2} (e^{j(\pi/4)n} + e^{-j(\pi/4)n}) = \frac{1}{2} e^{+j\Omega_o n} + \frac{1}{2} e^{-j\Omega_o n}$$

So fourier coefficients for $x[n]$ are

$$C_1 = \frac{1}{2}, C_{-1} = C_{-1+8} = C_7 = \frac{1}{2}$$

And all other $C_k = 0$ Choice (B)

26. As we know that Fourier transform of $h[n] = \frac{\sin \frac{\pi n}{8}}{\pi n}$ is

$$H(\Omega) = \begin{cases} 1 & |\Omega| \leq \pi/8 \\ 0 & \frac{\pi}{8} \leq |\Omega| \leq \pi \end{cases}$$

And $\Omega_o = \frac{2\pi}{N_o} = \frac{2\pi}{7}$ and filter passes only frequencies

in the range $|\Omega| \leq \frac{\pi}{8}$

So only D.C term will pass and discrete fourier series coefficient

$$C_o = \frac{1}{N_o} \sum_{n=0}^{N_o-1} x[n] = \frac{1}{7} \sum_{n=0}^6 1 = \frac{5}{7} \quad \text{Choice (C)}$$

27. From the figure $y[n] = x[n] + x[n-1]$

By taking Fourier transform $Y(\Omega) = X(\Omega) + e^{-j\Omega} X(\Omega)$

$$\text{So } H(\Omega) = \frac{Y(\Omega)}{X(\Omega)} = 1 + e^{-j\Omega}$$

$$= e^{-j\frac{\Omega}{2}} \left(e^{j\frac{\Omega}{2}} + e^{-j\frac{\Omega}{2}} \right)$$

$$= 2e^{-j\frac{\Omega}{2}} \cos\left(\frac{\Omega}{2}\right), |\Omega| \leq \pi \quad \text{Choice (D)}$$

28. $x(t) = \sin(125\pi t)$

$$x[n] = \sin\left(\frac{125\pi n}{500}\right) = \sin\left(\frac{n\pi}{4}\right)$$

$$\text{and } H(\omega) = \frac{\frac{1}{2} - e^{-j\omega}}{1 - \frac{1}{2}e^{-j\omega}}$$

$$\begin{aligned} \text{Magnitude } |H(j\omega)|_{\omega=\frac{\pi}{4}} &= \frac{\left(\frac{1}{2} - \frac{1}{\sqrt{2}}\right) + \left(\frac{j}{\sqrt{2}}\right)}{1 - \frac{1}{2\sqrt{2}} + \frac{j}{2\sqrt{2}}} \\ &= \frac{(\sqrt{2}-2+2j)}{(2\sqrt{2}-1)+j} \approx 1 \end{aligned}$$

$$< H(j\omega) = \tan^{-1}\left(\frac{2}{-2+\sqrt{2}}\right) - \tan^{-1}\left(\frac{1}{2\sqrt{2}-1}\right)$$

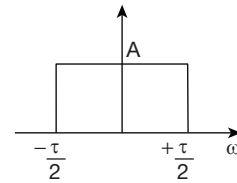
So, output of discrete filter $H(j\omega)$ is

$$y[n] = \sin\left(n\frac{\pi}{4} - 102.34^\circ\right) = -73.67 - 28.67 = -102.34^\circ$$

Choice (A)

29. $x(t) = \frac{\sin 2t}{2\pi t} \cdot \frac{\sin t}{2\pi t}$ & As we know that

$$\frac{A\tau}{2\pi} \frac{\sin\left(\frac{\tau}{2t}\right)}{\left(\frac{\tau}{2t}\right)} \underline{F.T.}$$



$$\text{So by comparing I function } \frac{\sin 2t}{2\pi t} = \frac{A\tau}{2\pi} \frac{\sin \frac{\tau}{2}t}{\left(\frac{\tau}{2}t\right)}$$

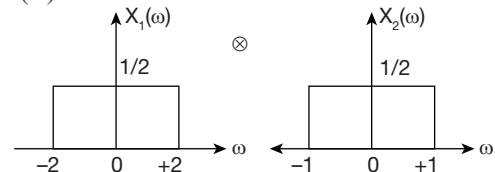
$$2t = \frac{\tau}{2}t; \Rightarrow \frac{\tau}{2} = 2; \tau = 4 \text{ and } \frac{A\tau}{2\pi} = \frac{1}{\pi}$$

$$\Rightarrow A = \frac{1}{2}$$

$$\text{By comparing II function } \frac{\sin t}{2\pi t} = \frac{A\tau}{2\pi} \frac{\sin \frac{\tau}{2}t}{\left(\frac{\tau}{2}t\right)}$$

$$\frac{\tau}{2} = 1 \Rightarrow \tau = 2 \text{ and } \frac{A\tau}{2\pi} = \frac{1}{2\pi} \Rightarrow A = \frac{1}{2}$$

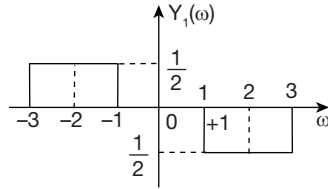
So $X(\omega)$



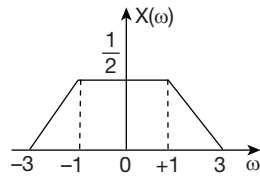
$$\text{Let } Y_1(\omega) = \frac{dX_1(\omega)}{d\omega} \cdot X_2(\omega)$$

$$= [\delta(\omega + 2) - \delta(\omega - 2)] \cdot X_2(\omega)$$

$$Y_1(\omega) = [X_2(\omega + 2) - X_2(\omega - 2)]$$



$$X(\omega) = \int y_1(\omega) d\omega$$



Choice (C)

30. Given that $x[n] = u[n] - u[n - 5]$
Even part of signal $x[n] = x_e[n]$

$$\text{So } x_e[n] = \frac{x[n] + x[-n]}{2}$$

$$x[-n] = u[-n] - u[-n + 5]$$

$$x_e[n] = \frac{u[n] + u[-n] - u[n - 5] - u[-n + 5]}{2}$$

$$= \frac{1 + \delta[n] - u[n - 5] - u[-n + 5]}{2}$$

$$= \frac{1 - u[n - 5] - u[-n + 5] + \delta[n]}{2}$$

$$= \frac{\{1 - u[-n + 5]\} - u[n - 5] + \delta[n]}{2}$$

$$= \frac{u[n + 4] - u[n - 5] + \delta[n]}{2}$$

Choice (C)

$$31. x(z) = \frac{z^{-1}(1 - z^{-4})}{(1 - z^{-1})^2(1 + z^{-1})} = \frac{z^{-5}(z^4 - 1)}{z^{-3}(1 - z)^2(z + 1)}$$

$$= \frac{1(z^2 + 1)(z - 1)z + 1}{z^2(z - 1)^2(z + 1)} = \frac{(z^2 + 1)}{z^2(z - 1)}$$

By Applying final value theorem

$$\lim_{n \rightarrow \infty} x[n] = \lim_{z \rightarrow 1} (z - 1)X(z)$$

$$= \lim_{z \rightarrow 1} \frac{(z - 1)(z^2 + 1)}{z^2(z - 1)} = \frac{(1 + 1)}{1} = 2$$

Choice (C)

32. Since $x(t)$ is real & odd, so $x[k]$ will be always odd imaginary.

Since $x(t)$ is real $x^*(t) = x(t)$

Again if $x(t)$ is an odd signal

So $x(t) = -x(-t)$

$$X[k] = -X[-k]$$

$$X^*[-k] = -X[-k]$$

for real & odd signal & also $x[0] = 0$ for an odd signal because in odd signal there must be zero at $k = 0$

Choice (A)

33. Since $X[k] = 0$ for $|k| > 1$

$$\& X[0] = 0$$

So Fourier series coefficient will present only at $k = 1$ & -1 & By parseval's theorem

$$\frac{1}{T} \int_0^T |x(t)|^2 dt = \sum_{k=-\infty}^{+\infty} |X[k]|^2$$

$$\frac{1}{2} \int_0^2 |x(t)|^2 dt = \sum_{k=-1}^{+1} |X[k]|^2 = 1$$

Since Time period = 2

$$\text{So } |X[-1]|^2 + |X[0]|^2 + |X[1]|^2 = 1$$

$$2|X[-1]|^2 = 1$$

$$|X[-1]|^2 = \frac{1}{2} \left(\begin{array}{l} \because |X[1]| = |X[-1]| \\ \& X(0) = 0 \end{array} \right)$$

fourier series coefficient is imaginary & odd

$$\text{So } X[1] = \frac{j}{\sqrt{2}}$$

$$X[-1] = \frac{-j}{\sqrt{2}}$$

$$\text{so } x(t) = \sum_{k=-1}^{+1} X[k] e^{jk\omega_0 t} \left(\because \omega_0 = \frac{2\pi}{2} = \pi \right)$$

$$= \frac{j}{\sqrt{2}} [e^{j\pi t} - e^{-j\pi t}] = \frac{2j \cdot j}{\sqrt{2}} \left[\frac{e^{j\pi t} - e^{-j\pi t}}{2j} \right]$$

$$= -\sqrt{2} \sin(\pi t)$$

Choice (A)

34. Since two systems are cascade so over all transfer

$$H(s) = H_1(s) H_2(s)$$

$$= \frac{1}{s+3} \cdot \frac{3}{(s+1)} = \frac{3}{2(s+1)} - \frac{3}{2(s+3)}$$

$$\text{So } h(t) = \frac{3}{2} [e^{-t} - e^{-3t}] u(t)$$

Choice (C)

35. for stability $\int_{-\infty}^{+\infty} h(t) dt < \infty$

$$\text{so } \int_0^{+\infty} \frac{3}{2} (e^{-t} - e^{-3t}) dt = \frac{3}{2} \left[\frac{e^{-t}}{-1} - \frac{e^{-3t}}{-3} \right]_0^{\infty}$$

$$= \frac{3}{2} \left[(0 - 0) - \left(-1 + \frac{1}{3} \right) \right] = 1$$

So overall system is stable and causal also because it depends upon past & present values.

Choice (D)